

RESEARCH INTERESTS

Resource-efficient deep learning, multimodal learning, medical and computational imaging, uncertainty quantification

EDUCATION

Cornell University

B.A. in Computer Science and Mathematics; GPA: 3.96

Ithaca, NY

Expected Dec 2028

EXPERIENCE

Cornell Computational Imaging Lab

Undergraduate Researcher

Ithaca, NY

Oct 2025 – Present

- Researching algorithms for rapid, statistically reliable, and interpretable microscopy imaging of biological tissue using deep learning and uncertainty quantification methods
- Exploring conformal prediction, bayesian inference, evidential deep learning, and designing transformer variants robust to variable acquisition conditions
- Contributed to new uncertainty calibration and quantile-conditioned U-Net methods (under review for conference)

Regeneron Pharmaceuticals

Quantum Computing Intern

Tarrytown, NY

Jun 2025 – Aug 2025

- Built quantum optimization code for protein–ligand docking with implications for more rapid drug development
- Modeled docking as a max-weighted clique problem, translated to quadratic unconstrained binary optimization, solved using quantum approximate optimization algorithm while benchmarking against classical brute-force approaches
- Presented work to internal Regeneron and IBM research teams, open-sourced the code; advised by the Chief Data Officer of the Regeneron Genetics Center

New Jersey Academy of Sciences

Software Engineer

Remote

Mar 2025 – May 2025

- Designed and operated a science fair judging program for the NJAS research symposium
- Served approximately 200 students and 80 judges, significantly reducing processing time and error compared to manual verification; software was subsequently refined and open-sourced for future, public use
- Helped team of 10 volunteers and event organizers to process and interpret results from the program

Montclair State University

Research Intern

Remote

Jan 2023 – Nov 2023

- Developed computer vision model for detecting toxin-producing cyanobacteria, improving on SOTA methods by more than 10% (avg. precision); attended IEEE conference and paper published in proceedings
- Handled data collection, uniting local data from dept. of environmental science with online iNaturalist data
- Implemented several optimizations such as genetic hyperparameter optimization and specialized data augmentations

Inventurn

Founder & Core Backend Developer

Remote

May 2022 – Nov 2024

- Built applications for businesses and nonprofits, including an outreach app serving thousands of volunteers in NJ
- Focused on APIs and database logic; also gained experience with web3 and blockchain infrastructure

CONFERENCES & JOURNALS

Li, B., Ding, K., Dera, D. (2025). MD-SA2: optimizing Segment Anything 2 for multimodal, depth-aware brain tumor segmentation in sub-Saharan populations. *J. Med. Imag.* 12(2). doi:10.1117/1.JMI.12.2.024007

Li, B., Serrano, K., Mazzaro, M., Wu, M., Wang, W., & Zhu, M. (2023). Identification of Cyanobacteria for Harmful Algal Blooms Research Using the YOLO Framework. *IEEE UEMCON*. doi:10.1109/uemcon59035.2023.10316078

HONORS

Regeneron Science Talent Search Top 40 Finalist (2025): The “oldest and most prestigious” science competition in the U.S. Awarded \$25,000 for brain tumor detection research with low-quality MRI scans from Sub-Saharan Africa.

1st Place Math & CS, NJ Academy of Sciences Symposium (2024): Placed first among approximately 200 students; represented NJ at national symposium. Research was on few-shot learning for tumor segmentation.

Karen Kranz Independent Researcher Award (2024): One of two recipients at Terra North Jersey Science Fair out of 150+ participants.

IEEE UEMCON Best Paper, AI/ML (2023): Recognized for research on toxin-producing algal bloom detection.

NJ Representative, JSHS (2023): 2nd in category-wide poster presentations at regional fair (over a dozen participating schools), advancing to national symposium. Research was on predicting viral zoonoses with genomic sequences.

Presidential Volunteer Service Award (Gold) (2022): For leading cleanup efforts at Liberty State Park, creating apps for volunteer organizations, and teaching Python.

VEX VRC Robotics World Championship Qualifier (2022): Programmer for team 7405M; implemented PID controller and odometry for autonomous routines in C/C++.

PROJECTS

Evidential Learning for Image Reconstruction

- Implemented novel uncertainty quantification via evidential deep learning for grayscale-to-RGB reconstruction, leveraging special normal inverse gamma convolutional layer.
- Wrote a related article, “Introduction to Deep Evidential Regression for Uncertainty Quantification”, that was featured as editor’s pick in *Towards Data Science* (publication with 150,000+ monthly subscribers).

Autograd from Scratch

- Led a team of four (Cornell Data Science) to build a vectorized reverse-mode automatic differentiation engine extending Karpathy’s micrograd.
- Supports additional operations relative to micrograd (e.g. log, softmax, matrix multiplications), SGD optimizer, MNIST validation (98% accuracy on test dataset), and a computation graph visualizer.

LLM Steganography

- Led Cornell Data Science ML engineers on building a system for detecting steganography (transmission of secrets within otherwise inconspicuous messages) from misspecified reward signals in LLMs via a sender/receiver/overseer adversarial setup.
- Worked to replicate NeurIPS paper “Hidden in Plain Text: Emergence & Mitigation of Steganographic Collusion in LLMs”; extended research by adding group relative policy optimization (GRPO) in place of PPO and fine-tuning the overseer probe.

Learning Based Cache Eviction

- Built synthetic workloads and data processing pipeline; trained ML-based cache eviction policy that performs pairwise judging using key usage patterns for improved data retrieval efficiency.
- Resulting model achieves up to 8% hit rate improvement over least recently used (LRU) heuristic baseline.

SKILLS

Languages: Python, Java, JavaScript, C/C++, SQL, LaTeX

Libraries: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, Qiskit

Tools: Linux/Unix, Git, Docker